Introduction

Mean reversion one of many type of quant strategy, which consider that if price have stationarity, standard deviation have a property to return back. Here I want to present first try of me using this concepts on practice. I want to test, if american indeces have a property to return back to VWAP from second standard deviation.

Formulas

VWAP (Volume Weighted Average Price)

Overview

VWAP is a technical analysis indicator that gives the average price a security has traded at throughout the session, based on both **price** and **volume**. It is heavily used by institutional traders as an execution benchmark.

The Formula

$$VWAP = \frac{\sum(TypicalPrice \times Volume)}{\sum Volume}$$

Where **Typical Price** for each period is calculated as:

$$TypicalPrice = \frac{High + Low + Close}{3}$$

Key Components Breakdown

- **Typical Price (TP):** Captures the full price action of a candle/period rather than just the close.
- **Volume (V):** The number of shares or contracts traded during that specific period.
- **Cumulative Sum ($\sum$):** Unlike a moving average, standard session VWAP resets at the open of every new trading day, accumulating data sequentially from session open.

VWAP Standard Deviation Bands

To measure price stretch and volatility around the baseline, standard deviation (σ) is calculated using volume-weighted variance:

Standard Deviation Formula

$$\sigma = \sqrt{\frac{\sum \left(Volume \times (TypicalPrice - VWAP)^2 \right)}{\sum Volume}}$$

Band Calculations

Upper and lower bands are plotted by multiplying the standard deviation by a factor (k, typically set to 1, 2, or 3):

$$UpperBand = VWAP + (k \times \sigma)$$

$$LowerBand = VWAP - (k \times \sigma)$$

Practical Trading Context

- **First Band ($\pm 1\sigma$):** Contains roughly 68% of session volume distribution; normal price fluctuation.
- **Outer Bands ($\pm 2\sigma$ or $\pm 3\sigma$):** Highlights overextended price action ("stretch"). In strong trends, prices can "ride" or "walk" along these outer bands rather than immediately reversing.

Rules to trade

My first goal is to find if price touches 2σ is price have a property to return to out VWAP. I check by horizons (5, 10, 15) bars, if we can return a profit make short and long events.